Artificial Intelligence (PEC-IT501B)

Course Details

Credit Points: 3

Syllabus Outline

Unit 1

Introduction. Overview of Artificial intelligence- Problems of AI, AI technique, Tic Tac Toe problem. Intelligent Agents. Agents & environment, nature of environment, structure of agents, goal based agents, utility based agents, learning agents. Problem Solving. Problems, Problem Space & search: Defining the problem as state space search, production system, problem characteristics, issues in the design of search programs.

Unit 2

Search techniques. Solving problems by searching problem solving agents, searching for solutions; uniform search strategies: breadth first search, depth first search, depth limited search, bidirectional search, comparing uniform search strategies. Heuristic search strategies. Greedy best-first search, A* search, memory bounded heuristic search: local search algorithms & optimization problems: Hill climbing search, simulated annealing search, local beam search, genetic algorithms; constraint satisfaction problems, local search for constraint satisfaction problems. Adversarial search. Games, optimal decisions & strategies in games, the minimax search procedure, alpha-beta pruning, additional refinements, iterative deepening.

Unit 3

Knowledge & reasoning. Knowledge representation issues, representation & mapping, approaches to knowledge representation, issues in knowledge representation. Using predicate logic. Representing simple fact in logic, representing instant & ISA relationship, computable functions & predicates, resolution, natural deduction.

Unit 4

Probabilistic reasoning. Representing knowledge in an uncertain domain, the semantics of Bayesian networks, Dempster-Shafer theory, Fuzzy sets & fuzzy logics.

Unit 5

Natural Language processing. Introduction, Syntactic processing, semantic analysis, discourse & pragmatic processing. Learning. Forms of learning, inductive learning, learning decision trees, explanation based learning, learning using relevance information, neural net learning & genetic learning. Expert Systems. Representing and using domain knowledge, expert system shells, knowledge acquisition.